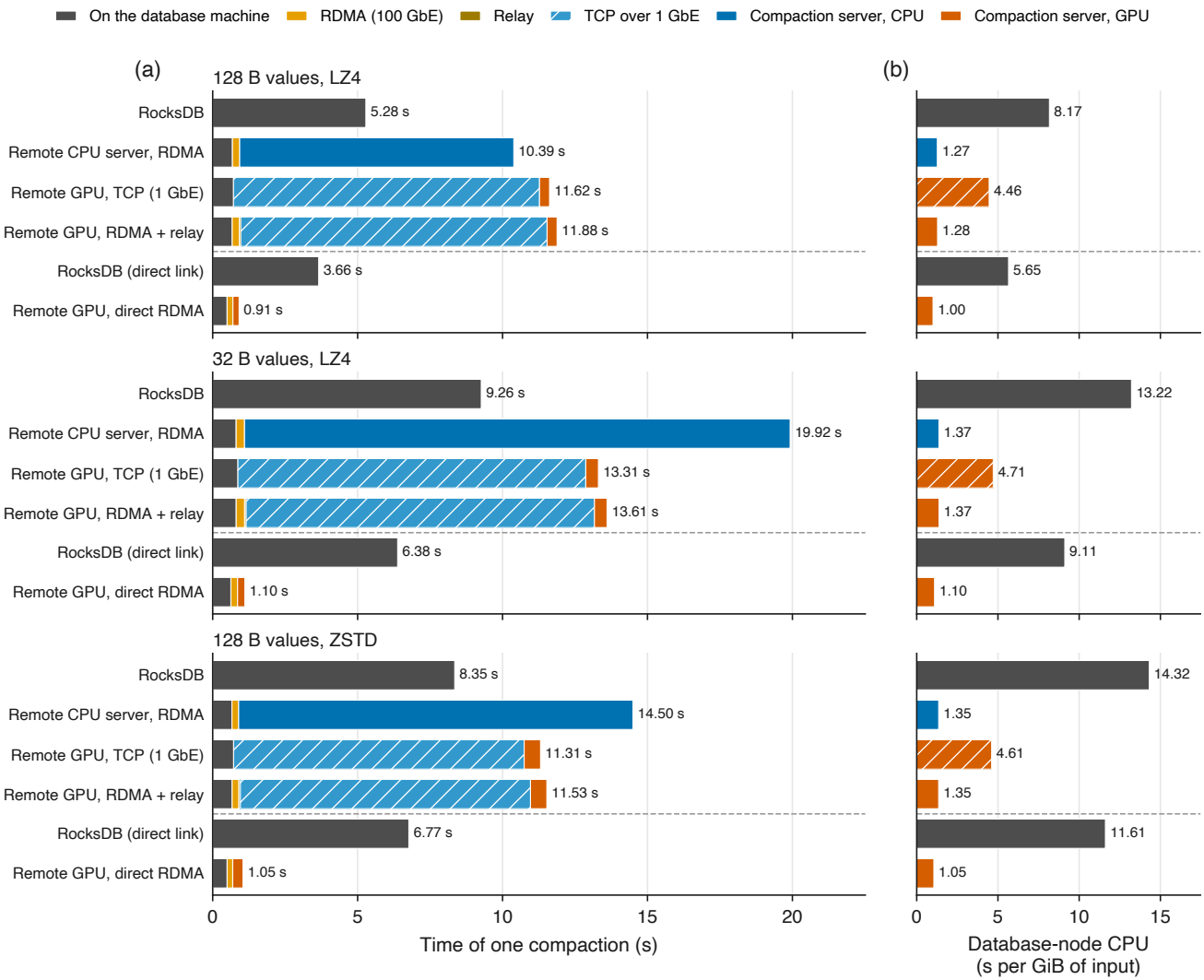
On the database machine
RDMA (100 GbE)
Relay
TCP over 1 GbE
Compaction server, CPU
Compaction server, GPU
(a)
(b)
128 B values, LZ4
RocksDB
Remote CPU server, RDMA
Remote GPU, TCP (1 GbE)
Remote GPU, RDMA + relay
RocksDB (direct link)
Remote GPU, direct RDMA
5.28 s
10.39 s
11.62 s
11.88 s
3.66 s
0.91 s
8.17
1.27
4.46
1.28
5.65
1.00
32 B values, LZ4
9.26 s
19.92 s
13.31 s
13.61 s
6.38 s
1.10 s
13.22
1.37
4.71
1.37
9.11
1.10
128 B values, ZSTD
8.35 s
14.50 s
11.31 s
11.53 s
6.77 s
1.05 s
14.32
1.35
4.61
1.35
11.61
1.05
0
5
10
15
20
Time of one compaction (s)
Database-node CPU
(s per GiB of input)